

ARCTIC BAY CS, NU, Canada

WMO: 715920

Lat: 72.993N

Lon: 85.012W

Elev: 10

StdP: 101.20

Time Zone: -6.00 (NAC)

Period: 00-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-40.1	-38.7	-45.0	0.0	-39.9	-43.6	0.1	-38.7	10.6	-21.4	9.4	-21.0	0.8	0	1.051

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.3	12.6	8.9	10.7	7.8	9.2	6.6	9.2	12.1	8.0	10.4	6.8	8.8	5.4	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.2	6.3	10.3	6.0	5.8	9.0	5.0	5.4	8.0	27.6	12.2	24.8	10.4	22.3	9.0	13.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.9	9.1	7.8	-42.2	15.6	1.5	2.3	-43.3	17.2	-44.2	18.5	-45.0	19.8	-46.1	21.4
			-42.3	10.9	1.5	1.3	-43.4	11.8	-44.2	12.6	-45.1	13.3	-46.2	14.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-12.7	-29.3	-31.8	-28.5	-18.9	-6.7	2.3	5.5	5.5	0.1	-7.8	-19.2	-25.2
	DBStd	14.38	5.11	5.11	5.75	5.38	4.77	2.51	2.34	2.64	2.86	5.19	5.87	5.75
	HDD10.0	8302	1217	1171	1194	866	519	232	143	143	298	553	875	1092
	HDD18.3	11339	1475	1404	1452	1116	777	482	399	399	548	811	1125	1351
	CDD10.0	4	0	0	0	0	0	0	2	2	0	0	0	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH29.1	0	0	0	0	0	0	0	0	0	0	0	0	0
(14) Wind	WSAvg	2.9	1.7	1.5	2.0	2.2	3.1	3.8	3.7	3.5	3.8	4.1	2.9	2.2
(15) Precipitation	PrecAvg	197	6	4	8	10	12	19	35	37	24	23	11	8
	PrecMax	302	12	12	28	28	35	42	76	66	54	57	23	19
	PrecMin	110	1	2	2	2	4	1	10	8	9	7	2	0
	PrecStd	45	3	3	6	6	7	11	17	15	12	12	6	4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-8.2	-16.9	-9.5	-3.8	3.8	10.4	15.4	15.2	7.6	3.2	-2.6	-7.3
		MCWB	-8.9	-17.5	-10.5	-4.9	1.7	6.4	10.0	11.2	5.3	1.4	-4.3	-8.3
	2%	DB	-16.4	-18.9	-13.5	-6.7	2.2	8.3	13.0	12.8	5.9	1.7	-6.3	-11.3
		MCWB	-17.1	-19.4	-14.0	-8.0	0.3	5.2	9.0	9.1	4.0	0.1	-7.5	-12.1
	5%	DB	-19.5	-21.0	-16.7	-8.6	0.9	6.8	11.0	11.0	4.8	0.2	-8.6	-13.9
		MCWB	-20.0	-21.4	-17.3	-9.6	-0.7	4.3	8.1	8.0	3.1	-1.1	-9.6	-14.4
	10%	DB	-22.0	-24.0	-20.3	-10.8	-0.3	5.6	9.4	9.5	3.9	-1.1	-10.8	-16.6
		MCWB	-22.4	-24.3	-20.8	-11.7	-1.6	3.5	6.8	7.0	2.4	-2.4	-11.6	-17.2
	0.4%	WB	-8.9	-17.5	-10.4	-4.9	1.9	7.0	10.6	12.0	5.6	1.7	-4.0	-8.1
		MCDB	-7.9	-17.1	-9.5	-3.9	3.8	9.8	13.6	14.6	7.4	2.8	-2.8	-7.3
	2%	WB	-17.0	-19.3	-14.0	-8.0	0.5	5.5	9.4	9.5	4.2	0.3	-7.5	-12.0
		MCDB	-16.4	-18.8	-13.4	-6.8	1.9	7.7	12.7	12.0	5.6	1.6	-6.4	-11.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	5%	WB	-19.9	-21.4	-17.2	-9.7	-0.6	4.5	8.1	8.2	3.2	-1.0	-9.6	-14.5
		MCDB	-19.5	-20.9	-16.6	-8.6	0.7	6.5	10.8	10.6	4.5	0.3	-8.7	-13.9
	10%	WB	-22.4	-24.3	-20.8	-11.7	-1.5	3.6	6.9	7.2	2.6	-2.3	-11.6	-17.2
		MCDB	-22.1	-24.0	-20.4	-10.8	-0.3	5.4	9.3	9.1	3.8	-1.1	-10.7	-16.7
	5% DB	MDBR	5.4	5.7	6.5	7.9	5.8	4.3	5.3	4.3	3.0	4.0	4.8	5.5
		MCDBR	7.8	8.3	9.3	8.4	5.8	6.3	8.4	7.0	3.7	4.4	6.6	9.1
(35) Mean Daily Temperature Range	5% WB	MCWBR	7.5	7.9	9.0	7.5	4.6	3.9	5.4	4.3	2.8	3.9	6.2	8.6
		MCDWR	7.8	8.3	9.2	8.3	5.6	6.1	8.0	6.5	3.6	4.0	6.5	8.9
	5% WB	MCWBR	7.5	7.9	8.9	7.4	4.6	4.0	5.5	4.3	2.8	3.7	6.1	8.5
		MCDWR	7.8	8.3	9.2	8.3	5.6	6.1	8.0	6.5	3.6	4.0	6.5	8.9
(40) Clear-Sky Solar Irradiance	taub		N/A	0.170	0.239	0.293	0.314	0.312	0.328	0.311	0.263	0.180	N/A	N/A
	taud		N/A	1.904	2.154	2.089	2.147	2.368	2.448	2.490	2.412	1.935	N/A	N/A
	Ebn at Noon		0	423	727	803	842	867	836	804	721	370	0	0
	Edh at Noon		0	27	64	103	114	96	86	71	52	24	0	0
(44) All-Sky Solar Radiation	RadAvg		0.00	0.16	1.50	4.12	6.19	6.31	4.96	3.21	1.29	0.31	0.00	0.00
	RadStd		0.00	0.02	0.10	0.20	0.36	0.69	0.48	0.40	0.20	0.05	0.00	0.00

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page